Emergency-Response Exercise

Confirming LI.FI's Ability to Protect User Funds

On 28 May 2026, LI.FI conducted a full end-to-end test of its automated emergency-pause capability in live production — and confirmed it works exactly as designed.

Prepared by the LI.FI Smart Contract Team

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For partners and integrators conducting security due diligence on LI.FI

Executive summary

LI.FI operates an automated safeguard that can **pause the LI.FI Diamond — its core cross-chain bridging and swapping protocol — within moments of a threat being detected**, freezing activity to protect user funds. The LI.FI Diamond is monitored around the clock by Hexagate, a Chainalysis company, and the emergency-pause system is designed so that this monitoring can trigger the response automatically, without waiting for a human at a keyboard.

On **28 May 2026** we tested that entire capability end-to-end, in production. The exercise was meticulously planned around a detailed written runbook, executed under a strict four-eyes principle, and deliberately staged so that it remained reversible at every step.

> **The result: the capability works.** Threat detection, authorization, the built-in safety stops, multi-channel team alerting, and controlled recovery all performed exactly as designed. As intended with any rigorous exercise, we identified a small set of operational refinements — and have since implemented all of them, further strengthening an already-working system.

Why this capability matters

The LI.FI Diamond moves real user funds across chains. If it ever came under attack, the most important defensive action is the ability to **pause it instantly** — freezing activity before an attacker can cause harm, and buying responders time to act.

A pause capability is only meaningful if it is **fast, reliable, and continuously ready**. It cannot depend on someone happening to be awake and online: it must be backed by continuous monitoring that can detect a threat and trigger the response automatically, and it must be proven under realistic conditions so there are no surprises when it matters most. This exercise was designed to demonstrate exactly that.

How the emergency response works

The response has **two distinct phases**. The first is automated and time-critical: it detects a threat and freezes the protocol within seconds, with no human in the loop. The second is deliberate and human-controlled: the protocol stays frozen while the team investigates, and normal operation is restored only once the team confirms it is safe to do so. Several independent checks must pass before the protocol is ever touched.

Two phases: an automated response that detects and freezes within seconds (blue), then a human-controlled recovery that follows investigation (outlined). The team is alerted in parallel.

Phase 1 — automated response (within seconds):

1. **Detection.** Hexagate continuously monitors the LI.FI Diamond for anomalous or malicious on-chain activity.
2. **Secured automation.** A detection is handed to a secured automation pipeline (built on GitHub Actions) that orchestrates the response. **Multiple independent authorization checks must pass before anything executes**, so no single component can trigger a pause on its own — a deliberate defense-in-depth design.
3. **Controlled pause.** Once the checks pass, the LI.FI Diamond is paused on-chain, freezing activity before an attacker can cause harm.

In parallel, the team is alerted across multiple channels — email, chat, and phone/SMS paging — so responders engage immediately, day or night.

Phase 2 — human-controlled recovery (on the team's timeline). A paused protocol stays frozen; **recovery is never automatic**. The pause holds while the team investigates and confirms it is safe to resume. Only then is normal operation restored, through a **multi-signature recovery that no single person can perform alone**. The recovery path is engineered and rehearsed ahead of time, so once the team authorizes it the protocol can be brought back promptly. This separation is deliberate: the freeze protects user funds for as long as needed, and the protocol returns to service only on an explicit human decision.

How we ensured a safe, controlled exercise

Testing an emergency control on a live system demands discipline. We applied several independent layers of control:

- **A four-eyes principle, executed as a team.** The entire smart contract team, together with a member of the security team, ran the exercise in a single shared session — working through every step of the runbook together rather than any one person acting alone. Each sensitive action was independently verified by a second team member before and after execution, so no critical step could be taken, or missed, by an individual.
- **A detailed written runbook.** Every action, its expected result, and each safety check was documented and reviewed in advance. The team executed against the runbook step by step rather than improvising.
- **A deliberately staged, two-stage design** to bound any impact. We first validated the detection and authorization chain in a mode where **no real pause was possible**, confirming the alarm and approval path worked end-to-end. Only then did we run a genuine pause — and only on a small set of deliberately chosen, low-traffic networks.
- **Pre-staged recovery for a reversible test.** Because this was a planned exercise of a deliberate pause — not a response to a real threat — we prepared and pre-signed the multi-signature recovery transactions in advance, purely so the test was reversible within minutes and required no scrambling. (In a real incident, recovery is signed and executed only after investigation; it is never pre-authorized.)
- **Advance notice to all stakeholders**, internal and external — including our external auditors — so the deliberately realistic alerts were never mistaken for a real incident.

What we tested and confirmed

Following the runbook, we exercised the full chain shown above and confirmed every stage performed as designed:

- **Detection and automated dispatch** — a Hexagate monitor detected a designated on-chain event and initiated the response automatically, with no human trigger, exercising the exact automated path a real threat detection would follow.
- **Authorization** — every independent authorization gate behaved correctly; no single component could act on its own.
- **Controlled execution** — the LI.FI Diamond was paused on-chain on the selected low-traffic networks.
- **Alerting** — the team was reached across every channel (email, chat, and phone/SMS paging), confirming responders would be engaged day or night.
- **Controlled recovery** — once the pause was confirmed, we restored normal operation within minutes using the multi-signature recovery transactions we had pre-signed for this test, confirming the deliberate, human-controlled exit path is fast and reliable when authorized. Everything returned to its pre-test state.

We also confirmed the system's **fail-safe design**: where a precondition was not met, the safeguard correctly **declined to act rather than acting incorrectly** — precisely the behaviour an emergency control must guarantee, and one of the most valuable assurances this exercise produced. Throughout, an internal readiness tool verified the exact state of the protocol on every network before, during, and after the exercise.

Continuous improvement

The core value of a controlled exercise is that it surfaces operational refinements while the stakes are low — so they are never encountered for the first time during a real incident. This test did exactly that, and **every improvement identified has since been implemented**:

Improvement	Status
Hardened funding checks across **all** production networks, with an automated guardrail ensuring the pause transaction is always funded.	✓ Implemented
Introduced an **automated weekly readiness check** that continuously verifies the emergency-pause system is correctly configured, funded, and ready.	✓ Implemented
Confirmed and hardened the direct on-chain execution path so the pause runs without delay.	✓ Implemented
Refined alerting so every alert is delivered as its own distinct incident, and reviewed on-call coverage so every responder is reachable.	✓ Implemented
Strengthened how operational secrets are validated, surfacing any issue well ahead of time.	✓ Implemented
Formalized that a security approver is present for the full duration of every live exercise, and that external auditors are notified in advance as a required step.	✓ Implemented

The exercise turned a previously one-time validation into an **ongoing, automated assurance process** — the protocol's emergency-pause readiness is now monitored continuously, not just at test time.

In closing

The most important question — *can LI.FI pause the LI.FI Diamond to protect user funds when it matters?* — is answered **yes**. The capability was tested end-to-end in live production, under disciplined controls, and confirmed working. The exercise both proved the core safeguard and made it measurably stronger. LI.FI treats the security of user funds as a continuous, evidence-based discipline, and this exercise is one part of that ongoing commitment.

Appendix

LI.FI Diamond

LI.FI's core cross-chain protocol: the on-chain smart contracts that power LI.FI's bridging and swapping, deployed across many networks. This report concerns the LI.FI Diamond specifically.

Emergency pause

An on-chain control that instantly freezes activity on the protocol to protect user funds during a suspected attack.

Four-eyes principle

A control requiring a second qualified person to independently verify each sensitive action, so no critical step is taken by one individual alone.

On-chain monitoring

Continuous, automated surveillance of smart-contract activity to detect threats in real time.

Controlled recovery

The deliberate, human-controlled lifting of a pause. A paused protocol stays frozen until the team has investigated and decided it is safe to resume; recovery is never automatic and is never pre-authorized. Lifting the pause requires approval from multiple independent signers (a multi-signature scheme), so no individual can resume the protocol alone.

About Hexagate. Hexagate, a Chainalysis company, is a real-time on-chain security platform that detects threats — exploits, key compromises, governance attacks, and phishing — and can trigger automated responses such as contract pauses. Built on Chainalysis's blockchain intelligence, it monitors the LI.FI Diamond continuously. Learn more: chainalysis.com/product/hexagate.